

WHERE BITS MATTER IN WORLD-MODEL PLANNING: A PAIRED MIXED-BIT STUDY FOR EFFICIENT SPATIAL REASONING

Anonymous authors

Paper under double-blind review

ABSTRACT

Efficient spatial reasoning requires world models that remain reliable under tight precision budgets. We study whether low-bit planning behavior is determined mostly by total bitwidth or by *where* bits are allocated across modules. Using DINO-WM on the Wall planning task, we run a paired-goal mixed-bit evaluation across uniform, mixed, asymmetric, and layerwise variants under two planner budgets. We observe a consistent three-regime pattern: 8/6-bit settings remain close to FP16, 3-bit settings collapse, and 4-bit settings are allocation-sensitive. In that transition region, preserving encoder precision improves planning relative to uniform quantization, and near-size asymmetric variants show the same encoder-side direction. These findings motivate module-aware, budget-aware quantization policies as a broader research direction for efficient spatial reasoning.

1 INTRODUCTION

World-model planning has become a strong paradigm for sample-efficient control and spatial reasoning, especially when latent dynamics are built on pretrained visual features (Hafner et al., 2023; Oquab et al., 2023; Zhou et al., 2024). Deployment constraints, however, force these models into tight memory and latency budgets where aggressive quantization can destabilize behavior. Recent broad surveys and world-model-specific studies report that low-bit degradation is often non-uniform across modules (Liu et al., 2025; Fu et al., 2026).

This paper asks a concrete workshop question: near low-bit operating points, is planning quality determined mainly by average precision, or by *bit allocation* across encoder and predictor? We study this in DINO-WM on Wall with paired-goal evaluation and mixed-bit variants. Our contribution is empirical but decision-relevant: we map where low-bit compression remains safe, where it fails, and which allocation choices are most promising under an explicit paired protocol. Concretely, we provide three takeaways: (1) a paired mixed-bit frontier showing an 8/6-safe, 4-transition, 3-collapse pattern; (2) robustness checks across two planner budgets plus goal-difficulty slices with consistent mixed-vs-uniform INT4 direction; and (3) encoder-focused evidence from asymmetric and layerwise ablations that sharpens future hypotheses.

2 RELATED WORK AND BROADER RESEARCH DIRECTION

Mixed-precision quantization has long argued that sensitivity is layer-dependent rather than uniform, with automated policies that optimize hardware-constrained objectives (Wang et al., 2018; Dong et al., 2019). Recent low-bit post-training quantization for foundation models further supports this module-sensitivity view: practical 4–8 bit deployment typically depends on how precision is allocated, which channels are protected, and how outliers are handled (Dettmers et al., 2022; Frantar et al., 2022; Xiao et al., 2022; Lin et al., 2023). Our results are consistent with that broader pattern, but in a planning setting where the metric is task success instead of perplexity or top-1 accuracy.

In world models, most efficiency work still emphasizes architecture or training design (Ha & Schmidhuber, 2018; Hafner et al., 2018; 2019; 2023), while quantization-specific evidence is only beginning to appear (Fu et al., 2026; Liu et al., 2025). The present study contributes a paired-evaluation template for this gap. The key observation is not that one specific mixed setting always

wins, but that a transition regime exists where placement of bits changes planning outcomes. For example, Table 3 shows that near-size encoder-heavy variants (E6/P4, E8/P4) improve over uniform INT4, while uniform INT6 demonstrates that higher total precision can still dominate under equal or slightly larger budgets.

This suggests a broader research direction for efficient spatial reasoning: treat bit allocation as a planning-aware resource-allocation problem, not a post-hoc compression detail. A scalable agenda is to optimize encoder/predictor precision jointly under explicit memory and latency constraints, then test whether the same transition structure holds across environments, checkpoints, and world-model families. If the structure persists, mixed-bit policies could become a first-class design axis for world-model deployment, analogous to horizon and optimizer budget choices.

3 SETUP

Task and metric. We use pretrained DINO-WM (Zhou et al., 2024) on Wall and keep the repository-native planning success metric from planning logs. We do not redefine task success. All variants use weight-only linear-layer quantization and identical checkpoints.

Model and hardware context. The model under study is DINO-WM with a visual encoder and latent predictor as defined in the original repository (Zhou et al., 2024). All experiments in this paper were executed locally on one Apple M4 MacBook Pro with 48GB unified memory. Runs use PyTorch with MPS when available and CPU fallback otherwise, under a single fixed software stack across all variants.

Quantization implementation details. We use post-training, weight-only quantization on `nn.Linear` layers via `experiments/quantize_utils.py`. For each output channel j and target bitwidth b , weights are quantized as $s_j = \max(|W_j|)/(2^{b-1} - 1)$ and $q_j = \text{clip}(\text{round}(W_j/s_j), -(2^{b-1} - 1), 2^{b-1} - 1)$, with dequantization $\tilde{W}_j = s_j q_j$ at inference. This is symmetric per-output-channel quantization with no activation quantization, no calibration pass, and no retraining. Quantized runs use the same `fake_int8` backend and FP16 runs use `fp16`; therefore, variant comparisons isolate allocation/bitwidth choices within a fixed inference path.

Variants. We evaluate FP16, uniform INT{8, 6, 4, 3}, mixed INT{8, 6, 4, 3} (encoder FP16, predictor quantized), asymmetric variants (E8/P4, E6/P4, E4/P8, E4/P6), and an INT4 layerwise encoder-retention sweep (0%, 25%, 50%, 75%, 100% encoder kept FP16; predictor fixed INT4).

Evaluation protocol. We use paired-goal evaluation: goals are generated once per seed and reused across variants. Budget bA uses (`goal_H = 9, opt_steps = 2, max_iter = 2`) with 3 seeds. Budget bB uses (12, 3, 3) with 2 seeds. Each run has 10 episodes, yielding 65 runs and 650 episodes total. The paired unit is (`seed, episode_id`): 30 paired units in bA and 20 in bB for the core mixed-vs-uniform comparisons. We estimate paired deltas with 4000-sample bootstrap confidence intervals at paired-unit granularity and report two-sided sign-test p values on non-tied pairs. We report success, model size, runtime, paired deltas, and log-derived mechanistic signals.

4 MAIN RESULTS

A three-regime pattern appears in this environment/checkpoint. Figure 1 and Table 1 show a stable regime at 8/6 bits, a transition at 4 bits, and collapse at 3 bits. At budget bA, FP16, uniform INT8, mixed INT8, and uniform INT6 all reach 0.533 success. At budget bB, FP16 and uniform INT6 both reach 0.650. In contrast, all 3-bit variants are 0.0 at both budgets.

4-bit is allocation-sensitive. At 4 bits, mixed INT4 is higher than uniform INT4: 0.267 vs 0.067 at bA, and 0.500 vs 0.200 at bB. Paired analysis gives +0.20 at bA (95% CI [0.00, 0.40]) and +0.30 at bB ([0.00, 0.55]), with sign-test $p = 0.109$ in both budgets. We therefore interpret the 4-bit effect as *directional* rather than statistically definitive (paired detail in Appendix Table 2 and forest view in Figure 8).

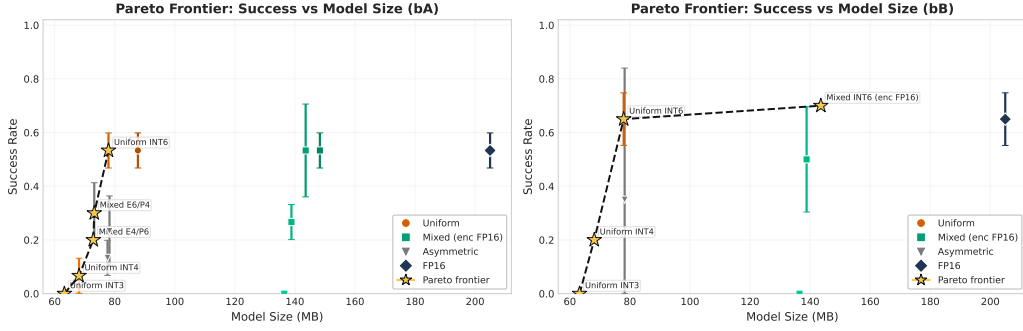


Figure 1: Success-size Pareto frontier for the paired mixed-bit study (budgets bA and bB). Each point is one variant, with vertical bars showing run-level 95% confidence intervals over seeds. Stars denote non-dominated Pareto points (higher success, lower model size). The frontier shows a stable 8/6-bit region, an allocation-sensitive 4-bit transition, and a collapsed 3-bit region.

Variant	Success bA	Success bB	Size (MB)
FP16	0.533	0.650	204.99
Uniform INT6	0.533	0.650	77.92
Mixed INT6	0.533	0.700	143.58
Uniform INT4	0.067	0.200	68.12
Mixed INT4	0.267	0.500	138.84
Uniform INT3	0.000	0.000	63.23
Mixed INT3	0.000	0.000	136.47

Table 1: Core mixed-bit outcomes from `my_mixedbit_run`. Means are shown; uncertainty and paired statistics are reported in Appendix Tables 2 and 3.

Fairness check: allocation versus total precision budget. Mixed INT4 (138.84 MB) is substantially larger than uniform INT4 (68.12 MB), so mixed-vs-uniform alone does not isolate allocation from total precision. To partially control this, we compare near-size asymmetric variants at bA: E6/P4 (73.19 MB, success 0.300) and E8/P4 (78.25 MB, 0.233) both improve over uniform INT4 (68.12 MB, 0.067), while remaining far smaller than mixed INT4. This suggests encoder-side bits are useful even under tight size budgets, while higher total precision (for example uniform INT6 at 77.92 MB, success 0.533) remains another strong driver (size-aware context in Appendix Table 3 and map in Figure 7).

Asymmetric and layerwise evidence supports encoder sensitivity. At bA, E6/P4 reaches 0.300 success at 73.19 MB, improving over uniform INT4 (0.067 at 68.12 MB), with paired delta +0.233 (95% CI [0.033, 0.433]). By contrast, increasing predictor bits while keeping encoder at INT4 (E4/P8, E4/P6) does not match mixed INT4. Numerically, E4/P8 is 0.133 with paired delta -0.133 relative to mixed INT4, suggesting that spending extra bits on the predictor cannot compensate for encoder degradation at this frontier. Layerwise INT4 ablation (Appendix Figure 3) is non-monotonic but peaks when the encoder is fully preserved (0.25 at 100% FP16), supporting the hypothesis that encoder precision is often a key bottleneck near the 4-bit frontier.

Mechanistic diagnostics are consistent with representation degradation. Across 65 run-level points (variant $\times$ budget $\times$ seed), success correlates negatively with divergence metrics from planning logs: Spearman $\rho = -0.928$ for mean state distance and $\rho = -0.708$ for visual-embedding divergence (Appendix Figure 5). This does not prove causality, but it supports a plausible mechanism: low-bit encoder degradation weakens latent geometry used for goal-directed planning.

5 DISCUSSION, LIMITATIONS, AND OUTLOOK

The data supports a narrower claim than “mixed is always better”: near low-bit transition points, planning quality depends on both allocation and total precision budget. The asymmetric results raise a practical design question for future work: can budget-constrained bit-allocation be optimized directly for planning success rather than model reconstruction error? The non-monotonic layerwise curve suggests a second question: which encoder blocks are precision-critical, and does that set change across environments? Finally, the strong divergence-success correlations motivate causal follow-ups, such as representation-preserving calibration or geometry-aware quantization objectives.

Within this study, generalization is operational rather than cross-domain. The same directional mixed-vs-uniform INT4 effect appears across two planner budgets and across goal-distance bins (Appendix Table 4), and encoder-priority trends recur across mixed, asymmetric, and layerwise families. These repeated patterns improve confidence that the 4-bit behavior is not tied to a single random goal draw.

This study is limited to one environment and one checkpoint family. Confidence intervals on key paired effects still include zero because the paired sample is modest. The headline mixed-vs-uniform comparison is also confounded by model size: mixed INT4 keeps an FP16 encoder and therefore uses substantially more total precision than uniform INT4. We evaluate weight-only quantization without activation quantization, retraining, or hardware-fused low-bit kernels. Due to compute constraints, all experiments were run on a single M4 MacBook Pro (48GB unified memory), which limits sweep breadth and runtime scale. As a result, this paper should be read as a strong preliminary study rather than a final benchmark. The next phase is to replicate and extend these experiments on NVIDIA GPUs, and to test broader quantization strategies including activation quantization, calibration-aware PTQ, and hardware-kernel-backed low-bit inference. Therefore, conclusions should be read as robust empirical hypotheses for efficient spatial reasoning, not universal laws.

6 CONCLUSION

In a paired mixed-bit study on DINO-WM planning, we find structured low-bit behavior: 8/6 bits are mostly safe, 3 bits collapse, and 4 bits is the sensitive transition regime. At 4 bits, mixed and asymmetric results consistently point toward encoder-side precision being more valuable than predictor-side precision, although effect sizes remain statistically limited and partially confounded by total precision budget. These findings motivate module-aware, budget-aware quantization policies as a concrete research direction for efficient spatial reasoning.

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A APPENDIX: EXTENDED RESULTS WITH CONTEXT

A.1 HOW TO READ THE APPENDIX

The main paper establishes the core claim from one frontier figure and one compact table. This appendix provides supporting analyses that answer three reviewer-facing questions: **(i)** Is the 4-bit effect robust to planner budget? **(ii)** Does evidence localize precision sensitivity to the encoder? **(iii)** Are observed gains consistent with a representation-degradation mechanism?

A.2 PAIRED MIXED-BIT EFFECT TABLE

Table 2 reports paired deltas for the most relevant comparisons. The first two rows are the primary claim (mixed INT4 vs uniform INT4 at bA and bB). The asymmetric rows test whether increasing encoder bits or predictor bits is more useful at similar model sizes.

Comparison	n pairs	Delta	95% CI	Sign-test p
bA: mixed_int4 - uniform_int4	30	+0.200	[0.000, 0.400]	0.109
bB: mixed_int4 - uniform_int4	20	+0.300	[0.000, 0.550]	0.109
bA: enc6_pred4 - uniform_int4	30	+0.233	[0.033, 0.433]	0.065
bA: enc4_pred8 - mixed_int4	30	-0.133	[-0.333, 0.067]	0.344

Table 2: Paired effect estimates from the consolidated mixed-bit pairwise statistics file.

Inference. Rows 1–2 show that mixed INT4 consistently outperforms uniform INT4 under paired episodes in both budgets, but with moderate uncertainty. Rows 3–4 strengthen the allocation interpretation: encoder-heavy E6/P4 improves over uniform INT4, while predictor-heavy E4/P8 does not recover mixed INT4. Together, the table supports an encoder-priority hypothesis near the 4-bit boundary.

A.3 SIZE-FAIRNESS AND ALLOCATION CONTEXT

Table 3 makes the size confound explicit. Mixed INT4 is much larger than uniform INT4, so we also compare near-size asymmetric variants. At bA, E6/P4 and E8/P4 are close to uniform INT4 in size and still improve success, indicating that encoder-side allocation helps even without fully preserving the encoder in FP16.

Variant (bA unless noted)	Success	Size (MB)	Note
Uniform INT4	0.067	68.12	baseline low-size 4-bit
E6/P4	0.300	73.19	near-size encoder-heavy
E8/P4	0.233	78.25	near-size encoder-heavy
Uniform INT6	0.533	77.92	higher total precision
Mixed INT4	0.267	138.84	larger precision budget
Uniform INT4 (bB)	0.200	68.12	budget-shift check
E8/P4 (bB)	0.350	78.25	near-size improvement
Mixed INT4 (bB)	0.500	138.84	larger precision budget

Table 3: Size-aware view of allocation effects. Mixed INT4 is stronger than uniform INT4 but uses substantially more total precision. Near-size asymmetric variants still show encoder-side gains over uniform INT4.

Inference. This table is the key fairness check for Figure 1. If the mixed INT4 gain were only due to larger size, near-size asymmetric variants would be expected to match uniform INT4. Instead, encoder-heavy near-size variants improve over uniform INT4, indicating that *where* bits are allocated matters in addition to total precision.

A.4 BUDGET ROBUSTNESS CONTEXT

Figure 2 shows that the mixed-vs-uniform INT4 gap persists when planner budget increases from bA to bB. This reduces the likelihood that the observed gap is only an artifact of one optimization budget.

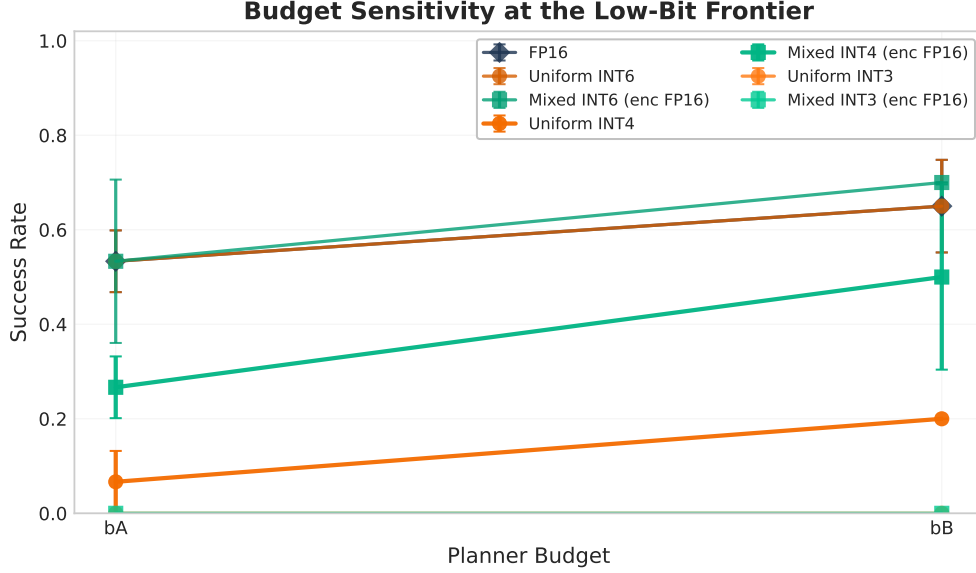


Figure 2: Budget robustness view. Uniform INT3 remains collapsed; mixed INT4 remains above uniform INT4 at both budgets.

Inference. The mixed-over-uniform INT4 direction survives when the planner budget increases from bA to bB. This reduces the chance that the headline finding is a single-budget artifact and supports treating the 4-bit transition as a structural regime.

A.5 ENCODER-LOCALIZATION CONTEXT

Figure 3 gives a coarse localization test: with predictor fixed at INT4, preserving more encoder precision improves peak success, although intermediate points are non-monotonic. This supports the main claim while motivating finer block-level studies.

Inference. Even in a coarse 5-point sweep, success rises toward the fully preserved encoder endpoint. Although non-monotonicity remains, the curve indicates that failures at 4-bit are not uniformly distributed across modules and that encoder precision is a dominant control variable.

A.6 DIFFICULTY-CONDITIONED CONTEXT

Figure 4 bins episodes by initial goal distance. Mixed INT4 is higher than uniform INT4 in most bins, suggesting the 4-bit allocation effect is not confined to a single difficulty slice.

Budget	Goal-distance bin	n	Uniform INT4	Mixed INT4
bA	low tertile	10	0.0	0.3
bA	mid tertile	10	0.0	0.3
bA	high tertile	10	0.2	0.2
bB	lower half	10	0.3	0.4
bB	upper half	10	0.1	0.6

Table 4: Difficulty-conditioned success from paired episodes. Directional mixed-vs-uniform gains appear across bins in both budgets.

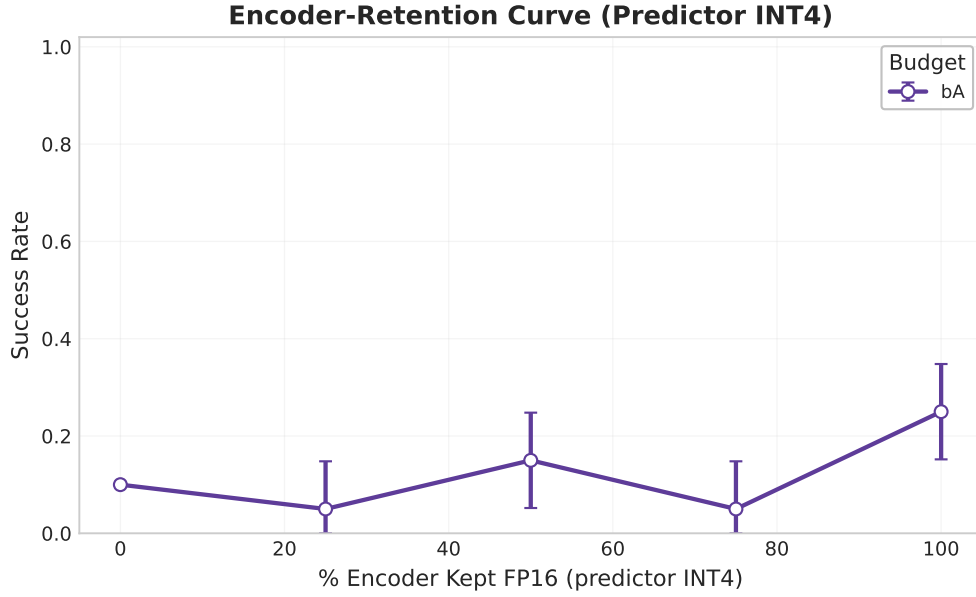


Figure 3: Encoder-retention sweep (INT4 predictor). The highest mean success occurs when encoder precision is fully preserved.

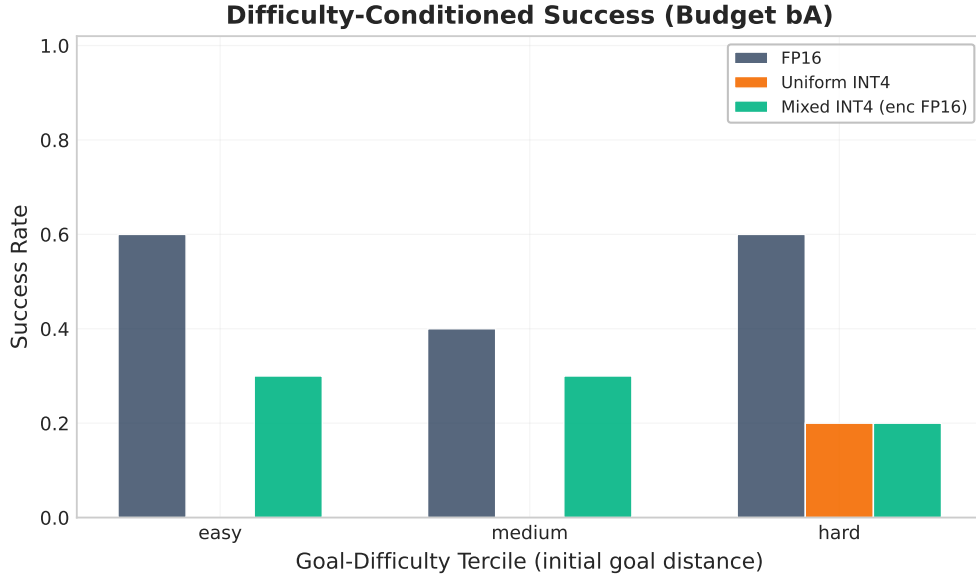


Figure 4: Difficulty-conditioned success at bA (paired episodes). Mixed INT4 is higher in most plotted bins.

Inference. The mixed-over-uniform direction appears in low and medium bins and does not reverse in hard bins. This suggests the allocation effect is not confined to one narrow difficulty regime, which improves confidence in transfer to broader evaluation sets.

A.7 MECHANISTIC CONTEXT

Figure 5 links variant-level success to divergence metrics extracted from planning logs. Correlations are computed over 65 run-level points (variant $\times$ budget $\times$ seed), with pooled budgets. The negative

trend supports a representation-degradation explanation for low-bit failure near the 4-bit boundary, while remaining correlational.

Mechanistic Signal: Embedding Divergence vs Success, Spearman $\rho=-0.71$

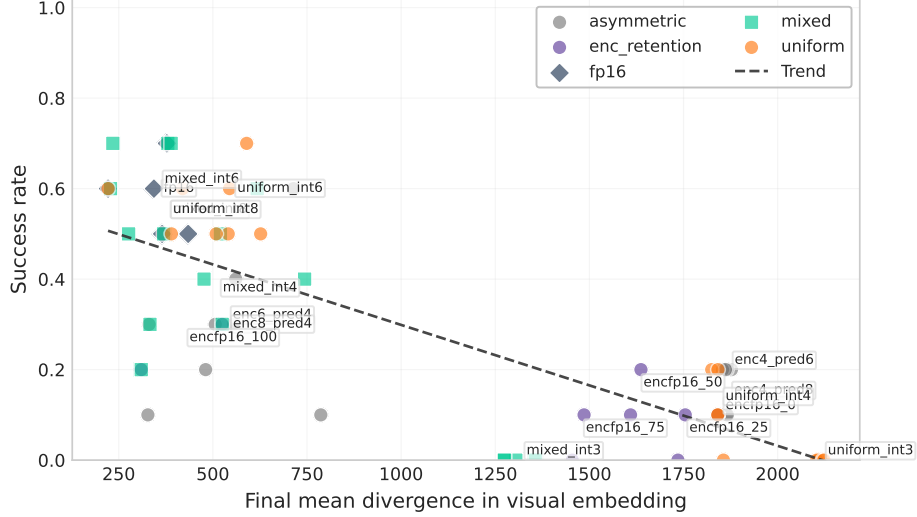


Figure 5: Mechanistic scatter: larger visual-embedding divergence is associated with lower planning success.

Inference. The negative trend links quantization behavior to representation quality rather than only planner noise. This does not establish causality, but it narrows follow-up experiments toward encoder-preserving or geometry-preserving quantization strategies.

A.8 MIXED-BIT MAPS FOR HYPOTHESIS GENERATION

The next two figures are exploratory and hypothesis-generating. Figure 6 visualizes the uniform/mixed bit ladder directly; Figure 7 maps asymmetric encoder/predictor allocations. Together they motivate a budgeted bit-allocation optimization problem for future work.

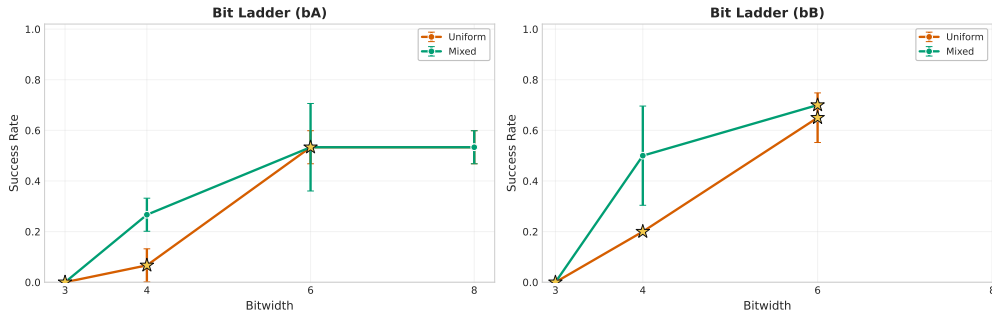


Figure 6: Uniform-vs-mixed bit ladder across budgets. Stars denote Pareto points in success-size space projected onto this ladder view.

Inference. The ladder view highlights a phase-transition-like structure: stable high-bit behavior, a sharp low-bit collapse, and a sensitive middle regime. The regime structure motivates adaptive bit-allocation policies instead of one-shot uniform compression.

Inference. At comparable predictor precision, moving encoder precision upward typically improves outcomes more than the reverse. Pareto-starred points concentrate in encoder-heavier configurations, reinforcing the encoder-priority conclusion from Tables 2 and 3.

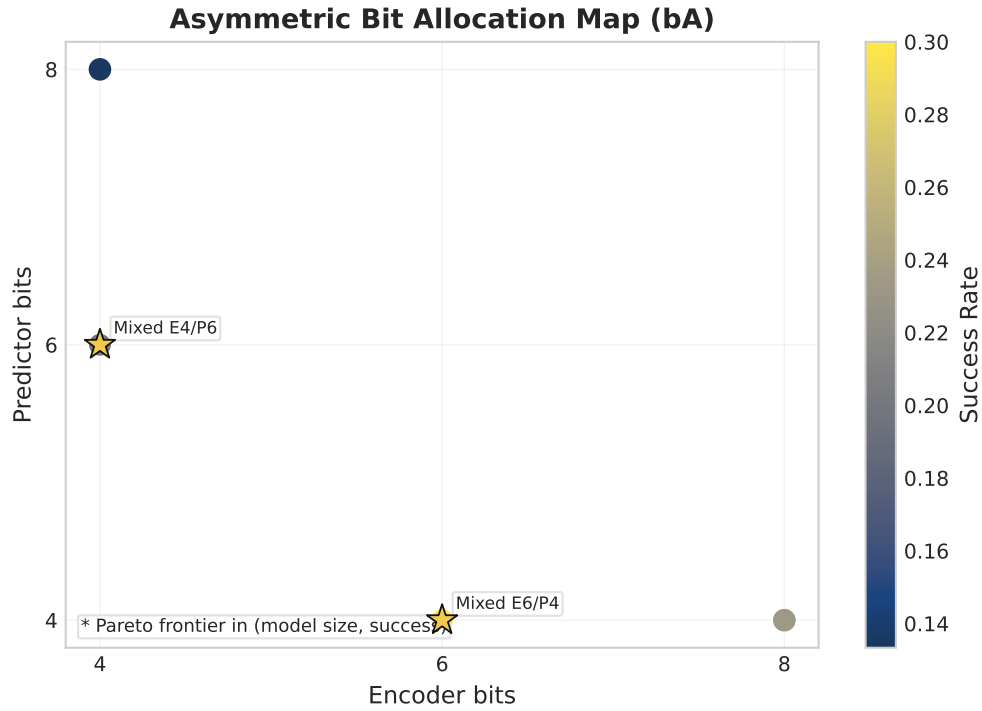


Figure 7: Asymmetric allocation map at bA. Marker color indicates success; star markers denote variants on the success-size Pareto frontier.

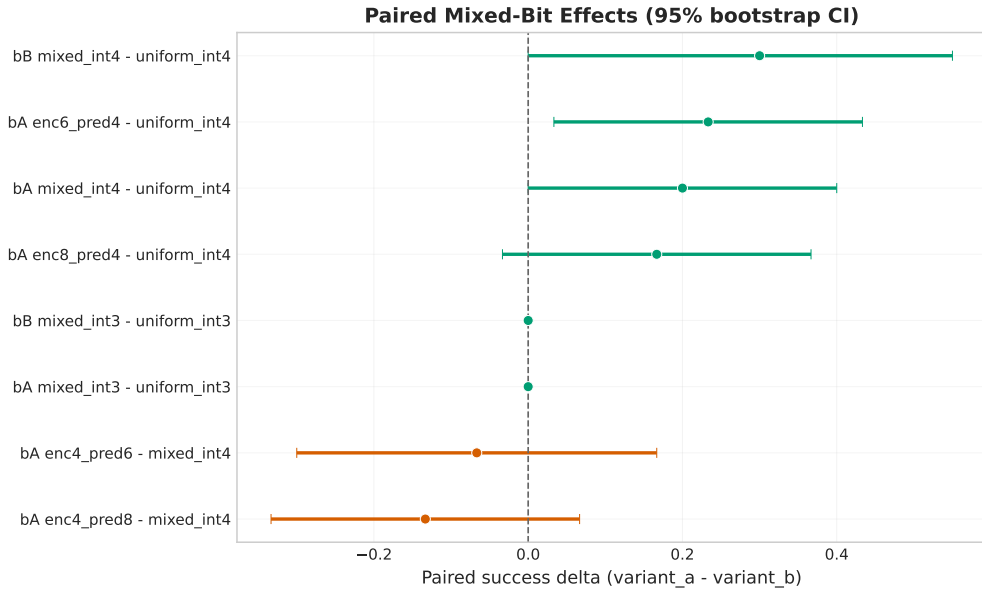


Figure 8: Forest plot of paired deltas with bootstrap confidence intervals for key comparisons.

Inference. The forest view makes effect direction and uncertainty transparent in one place. Positive deltas for mixed-vs-uniform INT4 and encoder-heavy-vs-uniform comparisons support the framing that allocation matters most in the transition regime.